Assignment 2 - VHDL Simulation and Synthesis

Task 1

Read the paper: Time and delta-delay in VHDL (found under Learning materials on Canvas).

Done :)

Explain in your own words what happens in a simulation cycle. You may illustrate your explanation with a flow diagram. Keep your answer short.

In a simulation cycle, to maintain concurrency, an event queue is created and sorted by the correct order of delta delays.

Explain what happens to signals that are assigned values with and without explicit delays.

Explicit vs implicit delay

1. Explicit delay - You specify after how much time the transition triggers
2. Implicit delay - It happens after a delta delay, often $10^{-15}s = 1\ \mathrm{fs}$

Task 2

Currently doing it :)

Task 3

How many flip-flops are used in the design?

There are 1370 flip-flops being used in this design.

How many LUTs are used in the design?

There are 284 LUTs being used in this design.

What is a LUT?

LUT = **L**ook **U**p **T**able

It is a tiny block of memory used to store the truth table of a logic function.

What is the total on-chip power?

The on-chip power is 0.403 Watts.

Mention things you can explore in the flowing steps of the design flow (Not sure exactly what is meant here):

Simulation

- Add and manage simulation sources (distinguishes between simulation and synthesis files), and configure simulation sets
- Run Behaviour Simulation on the RTL to check functionality
- Explore waveforms to see signals
- Verify design logic

Synthesis

- Elaborate RLT design (ex: explore RTL netlist, schematic view and so on)
- Browse the IP catalog (customise and generate IP blocks)
- Run RTL level **D**esign **R**ule **C**hecks (DRCs) to catch issues early
- After synthesis, one can explore:
    - Timing constraints and definition

- Different reports (ex. power, components)
- and so on :)

Implementation

- One can explore
  - Different reports (ex. timing, power ...), now based on actual placed and routed design
  - Actual routed connections on device
  - Generate the final bitstream file for programming the hardware device (FPGA)

$$\text{RTL} \rightarrow \text{Synthesis} \rightarrow \boxed{\text{I/O Planning}} \rightarrow \text{Placement} \rightarrow \text{Routing} \rightarrow \text{Bitstream}$$

Hand in answers to the questions

Will do!

Task 4

The files below are what were created.

`architecture.vhd`

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity latch is
    Port(A,B: in std_ulogic;
        Q, QN: inout std_ulogic := 'Z');
end latch;

architecture behavioral of latch is
begin
    Q <= A nor QN;
```

```
    QN <= B nor Q;
end behavioral;
```

`testbench.vhd`

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity test_bench is
end entity test_bench;

architecture test_latch of test_bench is
    signal A, B, Q, QN: std_ulogic;
    begin
        DUT: entity work.latch(behavioral)
        port map (A => A, B => B, Q => Q, QN => QN);
    stimulus: process is
        begin
            A <= '1' ; B <= '0' ;
            wait for 10 ns ;
            A <= '0' ;
            wait for 10 ns ;
            B <= '1' ;
            wait for 10 ns ;
            B <= '0' ;
            wait for 10 ns ;
            B <= '1' ; A <= '1' ;
            wait; --suspends the process so it does not loo
p :)
        end process stimulus;
end architecture test_latch;
```

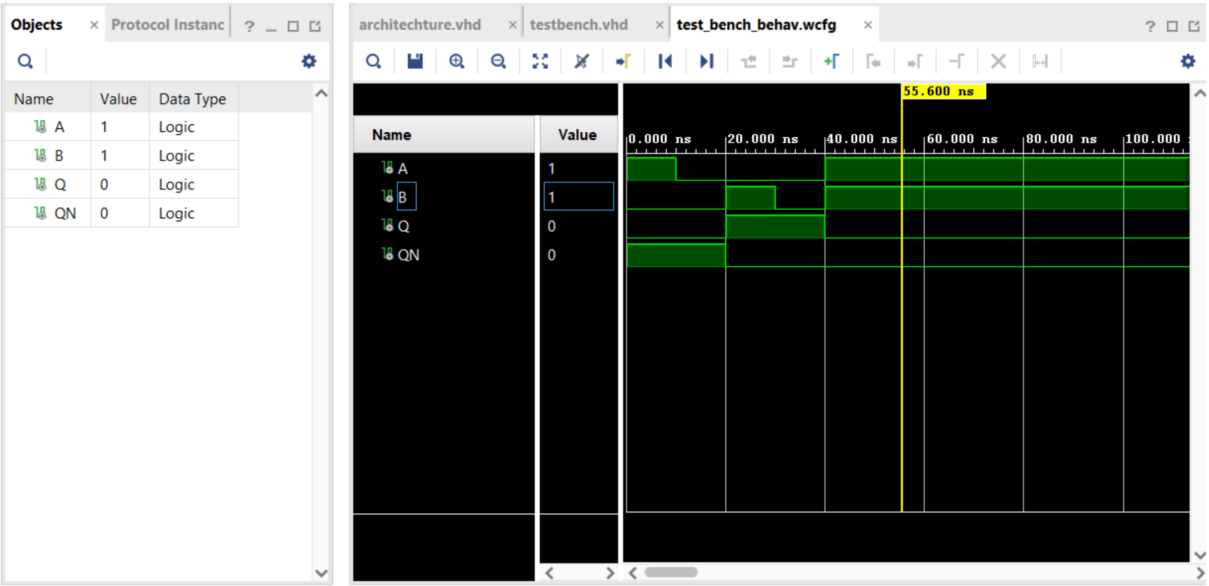
Simulation of the code

Does it work as expected?

Yes, seems to work. The output gives us what we expect.

Does it oscillate?

There is no oscillation. It gets stuck at 1100 (A B Q QN) after 40 ns.



What would happen if it ran the code sequentially

- When B = 1, after 30 ns, the Q and QN will both be calculated to be 0. This is because Q will use the older value instead of the updated value for QN. This should NOT happen in a latch.
- The rest of the tests would give the same results as in B.

Seeing delay

We only get errors ☹️ We didn't write in VERILOG, see picture, and we could not compile simulation tools required for Questa ☹️